

Duality is a structure; the sector is the cause

T-duality of the compact boson stated on its sectors and checked by two kernels, the measured invariant it predicts, and a criterion for when a self-dual point is physical

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SocrateAI-Scientific-QuantumFluids

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Abstract

An involution $R \leftrightarrow \alpha'/R$ on a length scale was, in earlier work of this programme, offered a physical home in T-duality, in K3 mirror symmetry, and in the modular symmetry of the quantum Hall system. Three rounds of pre-registered numerics and twenty-eight machine-checked theorems later, we can say where such a duality lives and what it does. We state T-duality on the sectors $(n, w) \in \mathbb{Z}^2$ of the compact boson and prove, in Lean 4 against Mathlib with two independent kernels, that (i) it is a reindexing of the sector lattice that leaves the sector sum invariant with no convergence hypothesis, (ii) the product of the electric and magnetic scaling dimensions is radius-independent, $\Delta_e \Delta_m = \frac{1}{4}$, which in the superfluid dictionary is $\eta n_s \lambda_T^2 = 1$ – the relation this programme measured to 12–27% on eleven energies in two boxes – and (iii) the self-dual radius $\sqrt{2}$ and the radius $2\sqrt{2}$ at which the vortex becomes marginal are different points: the Berezinskii–Kosterlitz–Thouless transition is the threshold of one sector, not the fixed point of the exchange. From these three facts and the literature on Ising, string-gas and superconductor–insulator dualities we extract a criterion: a self-dual point pins a transition only when the dual sectors are simultaneously critical on a discrete set of fixed points; on a line of fixed points it is a point like any other; when both sectors are populated it can be a preferred equilibrium. In every case what acts is a sector – a conserved label that crosses a threshold, proliferates, annihilates, or is imposed – and the duality relabels it. The duality is exact, verified, and not a cause. No physics here is new; the mathematics is elementary; what is new is that the distinction is stated so that a kernel can check it and an experiment can test it.

1 Three questions one involution raised

The dual-length involution $k \mapsto k_s^2/k$ of [1] has the form of T-duality. It was pointed out that the same involution is the Fricke element of $\Gamma_0(N)^+$, the mirror involution of degree-2N lattice-polarised K3 surfaces [2], and that $\Gamma_0(2)$ acts on the quantum Hall conductivity plane [3]. Each identification was taken to its gate [4, 5]: the Fricke matrix normalises Mathlib’s $\Gamma_0(n)$ and its axis action is the dual-length map, but no $\Gamma_0(n)$ acts on a wavenumber (`Fricke.lean`); the Fricke element fixes the Hall critical point’s orbit but sends every plateau off the odd-denominator class (`QHFricke.lean`); and the physical BKT point is not the self-dual radius. The residue of these three verdicts is one question: if a duality is exact and verified, and yet is not where the physics happens, what is it, and what is?

2 T-duality on the sector lattice, machine-checked

Let the compact boson have radius R in the normalisation where the self-dual radius is $\sqrt{2}$. The primary of momentum n and winding w has $h = \frac{1}{2}(n/R + wR/2)^2$, $\bar{h} = \frac{1}{2}(n/R - wR/2)^2$, hence

$$\Delta(n, w; R) = n^2/R^2 + w^2R^2/4, \quad s = h - \bar{h} = nw.$$

The electric (spin-wave, $e^{i\theta}$) operator is $(1, 0)$; the magnetic (vortex) operator is $(0, 1)$. The following are theorems of `CompactBoson.lean` (twelve statements, axiom footprint `{propext, Classical.choice, Quot.sound}` accepted by the Lean kernel and by `nanoda` through `Comparator`; three negative controls – product $\frac{1}{2}$, BKT at $\sqrt{2}$, duality with $R \mapsto 1/R$ – fail).

Theorem 1 (duality as reindexing). $\Delta(n, w; R) = \Delta(w, n; 2/R)$ and $s(n, w) = s(w, n)$. Consequently, for every weight F , $\sum_{(n, w) \in \mathbb{Z}^2} F(\Delta(n, w; R), s) = \sum_{(n, w) \in \mathbb{Z}^2} F(\Delta(n, w; 2/R), s)$.

The second statement is `Equiv.tsum_eq` applied to the swap of $\mathbb{Z}^2$: reindexing a sum by a bijection is an identity of `tsum` whether or not the series converges. This is what the duality is – a bijection of the sector lattice that preserves the spectrum. Savit’s review [6] gives the general form: a duality is a Fourier transform on the group of sectors, electric charges in G against magnetic charges in $\hat{G}$, with the Lagrangian sum over one family turned into the Hamiltonian sum over the dual family by Poisson summation.

Theorem 2 (the invariant). $\Delta(1, 0; R) \Delta(0, 1; R) = \frac{1}{4}$ for every $R \neq 0$.

In the XY / two-dimensional-superfluid dictionary $\Delta(1, 0) = \eta/2$ and $\Delta(0, 1) = n_s \lambda_T^2/2$, so the invariant reads $\eta n_s \lambda_T^2 = 1$. That is the relation measured in [5]: 1.047 and 1.072 on two equilibrated energies in round 2, 1.12–1.22 on the six energies of the heating ladder, 1.14–1.27 on the five energies of the $L = 32$ box – with the finite-size excess predicted by Hasenbusch’s logarithmic corrections [7]. *The numerical verification of the duality in this programme is the measurement of this product.*

Theorem 3 (self-dual is not marginal). For $R > 0$: $\Delta(1, 0) = \Delta(0, 1)$ iff $R = \sqrt{2}$, where both equal $\frac{1}{2}$; $\Delta(0, 1) = 2$ iff $R = 2\sqrt{2}$, where $\Delta(1, 0) = \frac{1}{8}$, i.e. $\eta = \frac{1}{4}$; and $2\sqrt{2} \neq \sqrt{2}$.

The transition is where the vortex operator becomes marginal – Nelson and Kosterlitz’s universal jump $n_s \lambda_T^2 = 4$ [8], located in our own runs at $T_{\text{BKT}} = 0.821$ with $\eta = 0.31$ there – and that is a condition on one sector. The exchange symmetry has its fixed point elsewhere, and nothing happens there.

3 When is a self-dual point physical?

The compact boson is the case with a *line* of fixed points, one for each R . Three other cases complete the criterion.

Ising: yes. Kramers–Wannier self-duality [9] fixes T_c at $\sinh 2K = 1$ because the order and disorder operators [10] exchange under the duality and $\mathbb{Z}_2$ has a single symmetric point: the dual sectors are simultaneously critical and the fixed-point structure is discrete.

String gas cosmology: a preferred equilibrium. In the Brandenberger–Vafa scenario [11, 12] “the symmetry we make use of is T-duality, and the new degrees of freedom are the string oscillatory modes and the string winding modes”. Momentum modes, with energy n/R , resist contraction; winding modes, with energy wR , resist expansion; when both are populated the radion is stabilised at the self-dual radius as the minimum of an effective potential. The self-dual point is physical there because both dual sectors are present and compete. And the exit from the Hagedorn phase – the event that makes three dimensions large – is the *annihilation of winding modes*: a sector event, with T-duality as the symmetry of the spectrum in which it is described.

Superconductor–insulator transition: a constraint. Boson–vortex duality [13, 14] predicts, if the transition is self-dual, a universal resistance $h/4e^2$ [15, 16]; measured values scatter around it. The cause of the transition is the proliferation of vortices on one side and of charges on the other; self-duality constrains the critical point only if the fixed point happens to be self-dual, which is a property of the fixed point, not of the duality.

Quantum Hall: organisation, not cause. $\Gamma_0(2)$ organises the plateau diagram and the position of the critical points [3]; the cause of the quantisation is Laughlin’s gauge argument [17] on the Chern sector; and the Fricke element, which fixes the critical orbit, is not a symmetry of the plateaux (`QHFricke.lean`).

Criterion. A self-dual point pins a transition iff the two dual sectors are simultaneously critical and the fixed-point set is discrete. On a line of fixed points the self-dual point is a point like any other and the transition sits where one sector crosses its own threshold. When both dual sectors are populated and compete energetically, the self-dual point is a preferred equilibrium. In each case the duality relabels the sectors and the sector acts.

4 What acts: the sector

The same triad recurs in every case above: a group of sectors ($\mathbb{Z}$ for $U(1)$ windings, $\mathbb{Z}_2$ for Ising, Chern numbers, string windings); a duality that is a Fourier or reindexing map between G -sectors and $\hat{G}$ -sectors preserving the spectrum; and physics that happens when a sector crosses a threshold, proliferates, annihilates, or is imposed. The causal role of the sector is what the rest of this programme established [5]: it is conserved by every continuous evolution without a phase slip (`TopologicalProtection`), additive in scale so that a bound pair is invisible from outside (`ScaleResolvedWinding`), read correctly by the sampled loop sum as the continuum degree (`ContinuumWinding`), it carries its own temperature (`SectorTemperature`), and in an energy-matched intervention it is a difference-maker: the same energy delivered as topology lowered the condensate by 0.49–0.73 where phonons lowered it by at most 0.035, with the injected sector reading five to eight times the bath’s temperature on our own calibrated thermometer. That is why one involution appears in helium, Hall bars, K3 moduli and string gases, and why it explains none of them.

5 What this paper does not claim

- Novelty in the mathematics: the three theorems are elementary; their value is that the distinction between the duality and the sector is stated so that a kernel checks it.
- Novelty in the physics: Kramers–Wannier, Nelson–Kosterlitz, Brandenberger–Vafa, Fisher and Lütken–Ross are cited, not extended.
- Any identification of a superfluid with a K3 surface or a string background. The K3 mirror involution is the same map on a different space; the criterion above is what the shared map does and does not license.
- That every duality in physics fits the triad: the cases examined do; the general statement is a reading.

Addendum (2026-09-26): the criterion sharpened by the non-invertible-symmetry literature

A literature review after publication located the modern machinery that makes §3’s criterion precise rather than descriptive. Aasen, Mong and Fendley state the core distinction directly: a self-duality is implemented by an operator that “is not a symmetry in the traditional sense... not unitary or even invertible” [18]. Choi, Córdova, Hsin, Lam and Shao exhibit the Poisson resummation that turns this into an explicit lattice statement, and prove that at some self-dual

points a mixed 't Hooft-like anomaly *forbids* a trivially gapped phase – an anomaly argument, not the duality map, is what does the forbidding [19]. Kaidi, Ohmori and Zheng give the clean mechanism for our own already-published observation: at $SO(3)$ Yang–Mills $\theta = \pm\pi$, the self-dual point sits *between two physically inequivalent gapped phases* (a $\mathbb{Z}_2$ TQFT and the trivial phase), which is why it is forced to be a transition [20]. Shao’s lecture notes give the general rule, in one sentence: away from a critical point Kramers–Wannier duality “is not a duality relating two equivalent descriptions of a single system... it is a map from the high-temperature phase to the low-temperature phase,” and at the compact boson’s self-dual radius $\tau = i$ “the duality defect reduces to an invertible $\mathbb{Z}_4$ symmetry” [21] – exactly why $R = \sqrt{2}$ (a point on a continuous moduli space with enhanced symmetry) is structurally unlike the Ising or $SO(3)$ -Yang–Mills self-dual points (between two gapped, inequivalent phases), and why neither is like the vortex-marginality radius $2\sqrt{2}$, which is a threshold on one sector alone. One further correction: Hayashi and Tanizaki show that the Cardy–Rabinovici $SL(2, \mathbb{Z})$ “self-duality” of the 2D Coulomb gas/ $\mathbb{Z}_N$ clock model is not naive – it holds only after gauging the electric one-form symmetry with a discrete θ -term – and that the resulting self-dual point, where three first-order transition lines meet, is again forced non-trivial by a separate mixed gravitational anomaly, not by the duality symmetry itself [22]. None of this changes a theorem of this paper; it replaces the numerical observation “ $\sqrt{2} \neq 2\sqrt{2}$ ” with an explicit criterion (inequivalent sectors on the two sides $\Rightarrow$ forced transition; a continuous moduli space with enhanced symmetry $\Rightarrow$ no transition) that a future module could formalise as a finite-group gauging statement.

Addendum (2026-09-26, part 2): two more cases, same criterion

Two further cross-domain instances, found by literature review and independently verified, apply the criterion of §3 without changing it.

RCFT/Verlinde: the sharp case, and the same verdict as BKT. The Verlinde formula [23], proved from Moore–Seiberg’s polynomial equations [24, 25], makes the modular S -matrix acting on the primaries (the sectors) of a rational conformal field theory a duality satisfying $S^2 = C = (ST)^3$ exactly, for S unitary and symmetric [26] – sharper than this programme’s own $\mathbb{Z}_N$ case, where Mathlib’s unnormalised `ZMod.dft` gives $\text{dft} \circ \text{dft} = N \cdot \Phi(-\cdot)$, not the identity (`SectorDuality.lean`). S is a literal finite-abelian-group Fourier transform only for simple-current (pointed) fusion rings – e.g. $\widehat{su}(2)_1$, whose two primaries fuse as $\mathbb{Z}_2$ and whose S is exactly the (unnormalised) Hadamard matrix $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ – and here $S^2 = C = \mathbb{K}$ with no extra factor once the missing $1/\sqrt{2}$ is restored (`RCFTDuality.su2Level1_S_sq`), the clean identity the unnormalised $\mathbb{Z}_N$ statement only approximates up to the decoupled factor $|G| = 2$. For a generic (non-abelian) fusion category S is instead the Kac–Peterson matrix and no group Fourier transform underlies it at all; this module proves only the 2×2 integer-matrix identity, not that it is the Kac–Peterson matrix, which is affine-Lie- algebra representation theory supplied here as physics input. The natural self-dual point of the worked example – the free-boson radius $R = 1$ where $\widehat{u}(1)$ enhances to $\widehat{su}(2)_1$ under T-duality $R \leftrightarrow 1/R$ – sits on a *continuous* line of RCFT fixed points, $1 \leq R \leq \sqrt{2}$, separating three gapped phases [27], not on a discrete pair of inequivalent sectors: by this programme’s own criterion it is a point like any other, not a forced transition – the Ising verdict does not transfer, the BKT verdict does.

Particle-vortex/level-rank in the fractional quantum Hall effect: a second, orthogonal duality. At $\nu = \frac{1}{2}$ the Halperin–Lee–Read composite fermion is conjectured to be a massless Dirac fermion, exactly self-conjugate under the particle-hole (CT) involution that exchanges the Jain-sequence pair $n/(2n+1) \leftrightarrow (n+1)/(2n+1)$ [28]; Seiberg–Senthil–Wang–Witten embed this in a fermionic particle-vortex duality web [29] whose rigorously-established topological limit is level-rank duality, $U(N)_K \leftrightarrow U(K)_N$ [30, 31]. This is a different duality from $\Gamma_0(2)$ on the conductivity

plane, acting on a different sector: the finite set of Chern–Simons/WZW primaries, labelled by Young diagrams in an $N \times K$ box, relabelled by transpose to a $K \times N$ box – an operation `Mathlib` already formalises on `YoungDiagram` as an order isomorphism (`transposeOrderIso`), restricted here to two finite boxes (`LevelRankDuality.levelRankRelabeling`). Filling $\nu = \frac{1}{2}$ is not itself a transition between two gapped, inequivalent sectors – that role belongs to the Jain pairs on either side of it – but the unique fixed point of the particle-hole involution on the continuous filling ν , in the same slot as the compact boson’s self-dual radius: an enhanced-symmetry point, not a cause. What self-duality there fixes is a number, not a phase boundary – the composite fermion’s Berry phase is forced to be exactly π and $\sigma_{xy}^{CF} = -\frac{1}{2}$ exactly [28] – so this instance sits with the superconductor–insulator transition in the *constraint* slot, not with $\Gamma_0(2)$ ’s *organisation* slot. Level-rank duality itself, wherever it appears in the non-Abelian FQH hierarchy, is unambiguously organisation-not-cause: a proved relabelling of a finite anyon label set that decides no physics and causes no transition. Neither Lean module proves the physical content of its own duality (S,T-matrix or Verlinde-formula preservation for level-rank; that the Hadamard matrix is the Kac–Peterson matrix for RCFT) – in both cases the module certifies a decidable combinatorial or linear-algebraic skeleton, and the identification with the named physical duality is supplied as literature-sourced prose, exactly as `SectorDuality.lean` and `QHFricke.lean` already do.

Addendum (2026-09-26, part 3): a ninth case, and one explicitly declined

A further literature pass, following the same criterion, adds one more continuous-window case and explicitly excludes a candidate that turned out, on reading, not to be a duality at all.

Seiberg duality: a third continuous window, same verdict. Electric $SU(N_c)$ SQCD with N_f flavors and its magnetic dual $SU(N_f - N_c)$ [32] flow to the same interacting fixed point throughout the conformal window $\frac{3}{2}N_c < N_f < 3N_c$ [33] – “completely equivalent at long distance,” not two phases exchanged by the duality. The sector here is the integer N_c itself, relabelled by $N_c \mapsto N_f - N_c$; the point $N_c = N_f/2$, where the two gauge groups coincide, is a direct algebraic fact about this map, not a distinguished physical locus – no transition occurs there, the whole window being one phase. Same verdict as the compact boson and the RCFT self-dual radius: a point on a continuum, not a cause. No Lean statement is proposed: the only formalisable content of the map $N_c \mapsto N_f - N_c$ is a one-line integer involution, and the actual physical duality (holomorphy, anomaly matching, RG-flow equivalence) has no combinatorial skeleton analogous to `YoungDiagram.transpose` or `ZMod.dft`; unlike Kramers–Wannier or T-duality, the two Lagrangians are not even operator images of each other (the magnetic side carries an extra gauge-singlet meson with its own superpotential).

Topological superconductors: not a duality. Reading Kitaev’s classification of unpaired Majorana modes [34], the Altland–Zirnbauer symmetry classes [35], and Ryu–Schnyder–Furusaki–Ludwig’s tenfold-way classification [36] directly finds no duality organising this material: it is a classification of gapped free-fermion phases by Clifford-algebra K-theory and Bott periodicity, related by dimensional reduction, not an equivalence between two theories. It has a sector (a $\mathbb{Z}$ or $\mathbb{Z}_2$ topological invariant) but no map that relabels it against a dual description – the word “duality” organises none of these three sources. This is therefore *excluded*, not merely deferred: forcing this programme’s duality/sector vocabulary onto a classification would be the formalising-an-analogy failure this programme’s own discipline exists to catch. One legitimate but pre-existing link is worth recording as a footnote rather than a new case: the 1D Kitaev chain (class BDI) is related by the Jordan–Wigner transform to the transverse-field Ising model, so its topological/trivial transition is, in fermionic language, the same Kramers–Wannier self-duality

already the Ising case above – not a ninth instance, a relabelling of the first one.

References

- [1] Xavier Callens. Formalizing quantum-fluid structure in Lean 4. Zenodo, 2026.
- [2] I. V. Dolgachev. Mirror symmetry for lattice polarized K3 surfaces, 1995. *J. Math. Sci.* 81, 2599 (1996).
- [3] C. A. Lütken and G. G. Ross. Duality in the quantum Hall system. *Phys. Rev. B*, 45:11837–11845, 1992.
- [4] Xavier Callens. How loose is Wasserstein stability on a vortex field? Zenodo, 2026.
- [5] Xavier Callens. Does topology causally influence physics? winding numbers as difference-makers in a two-dimensional Bose gas. Zenodo, 2026.
- [6] R. Savit. Duality in field theory and statistical systems. *Rev. Mod. Phys.*, 52:453–487, 1980.
- [7] M. Hasenbusch. The two dimensional XY model at the transition temperature: a high precision Monte Carlo study, 2005. *J. Phys. A* 38, 5869 (2005).
- [8] D. R. Nelson and J. M. Kosterlitz. Universal jump in the superfluid density of two-dimensional superfluids. *Phys. Rev. Lett.*, 39:1201–1205, 1977.
- [9] H. A. Kramers and G. H. Wannier. Statistics of the two-dimensional ferromagnet. Part I. *Phys. Rev.*, 60:252–262, 1941.
- [10] E. Fradkin and L. Susskind. Order and disorder in gauge systems and magnets. *Phys. Rev. D*, 17:2637–2658, 1978.
- [11] R. Brandenberger and C. Vafa. Superstrings in the early universe. *Nucl. Phys. B*, 316:391–410, 1989.
- [12] R. H. Brandenberger. String gas cosmology, 2008.
- [13] C. Dasgupta and B. I. Halperin. Phase transition in a lattice model of superconductivity. *Phys. Rev. Lett.*, 47:1556–1560, 1981.
- [14] M. P. A. Fisher and D. H. Lee. Correspondence between two-dimensional bosons and a bulk superconductor in a magnetic field. *Phys. Rev. B*, 39:2756–2759, 1989.
- [15] M. P. A. Fisher. Quantum phase transitions in disordered two-dimensional superconductors. *Phys. Rev. Lett.*, 65:923–926, 1990.
- [16] M.-C. Cha, M. P. A. Fisher, S. M. Girvin, M. Wallin, and A. P. Young. Universal conductivity of two-dimensional films at the superconductor-insulator transition. *Phys. Rev. B*, 44:6883–6902, 1991.
- [17] R. B. Laughlin. Quantized Hall conductivity in two dimensions. *Phys. Rev. B*, 23:5632–5633, 1981.
- [18] D. Aasen, R. S. K. Mong, and P. Fendley. Topological defects on the lattice: I. The Ising model. *J. Phys. A*, 49:354001, 2016.
- [19] Y. Choi, C. Córdova, P.-S. Hsin, H. T. Lam, and S.-H. Shao. Non-invertible duality defects in $3 + 1$ dimensions. *Phys. Rev. D*, 105:125016, 2022.

- [20] J. Kaidi, K. Ohmori, and Y. Zheng. Kramers–Wannier-like duality defects in $(3 + 1)$ d gauge theories. *Phys. Rev. Lett.*, 128:111601, 2022.
- [21] S.-H. Shao. What’s done cannot be undone: TASI lectures on non-invertible symmetry, 2023.
- [22] R. Hayashi and Y. Tanizaki. Non-invertible self-duality defects of Cardy–Rabinovici model and mixed gravitational anomaly, 2022.
- [23] E. Verlinde. Fusion rules and modular transformations in 2D conformal field theory. *Nucl. Phys. B*, 300:360–376, 1988.
- [24] G. Moore and N. Seiberg. Polynomial equations for rational conformal field theories. *Phys. Lett. B*, 212:451–460, 1988.
- [25] G. Moore and N. Seiberg. Classical and quantum conformal field theory. *Commun. Math. Phys.*, 123:177–254, 1989.
- [26] J. Fuchs. Fusion rules in conformal field theory, 1993.
- [27] S. D. Pace, A. Chatterjee, and S.-H. Shao. Lattice T-duality from non-invertible symmetries in quantum spin chains, 2024.
- [28] D. T. Son. Is the composite fermion a Dirac particle? *Phys. Rev. X*, 5:031027, 2015.
- [29] N. Seiberg, T. Senthil, C. Wang, and E. Witten. A duality web in $2+1$ dimensions and condensed matter physics. *Ann. Phys.*, 374:395–433, 2016.
- [30] P.-S. Hsin and N. Seiberg. Level/rank duality and Chern–Simons–matter theories. *JHEP*, 09:095, 2016.
- [31] S. G. Naculich and H. J. Schnitzer. Level-rank duality of the $U(N)$ WZW model, Chern–Simons theory, and 2d qYM theory. *JHEP*, 06:023, 2007.
- [32] N. Seiberg. Electric-magnetic duality in supersymmetric non-Abelian gauge theories. *Nucl. Phys. B*, 435:129–146, 1995.
- [33] K. Intriligator and N. Seiberg. Lectures on supersymmetric gauge theories and electric-magnetic duality, 1996.
- [34] A. Yu. Kitaev. Unpaired Majorana fermions in quantum wires. *Phys.-Usp.*, 44:131, 2001.
- [35] A. Altland and M. R. Zirnbauer. Nonstandard symmetry classes in mesoscopic normal-superconducting hybrid structures. *Phys. Rev. B*, 55:1142–1161, 1997.
- [36] S. Ryu, A. P. Schnyder, A. Furusaki, and A. W. W. Ludwig. Topological insulators and superconductors: tenfold way and dimensional hierarchy. *New J. Phys.*, 12:065010, 2010.